

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Bachelor of Science in Information Technology (B.Sc. IT) Semester-I University Examination

November-2019

Date: 13.11.19, Wednesday

Time: 10.00 a.m. to 1.00 p.m.

INFORMATION TECHNOLOGY AND DIGITAL ELECTRONICS [CA403]

Marks: 70

Duration: 180 mins.

SECTION-I

Answer all the questions.

Section Duration: 30 mins

Make suitable assumptions when necessary.

Simple Calculator is permitted.

- 1) X-ray is an example of _____ kind of data. (1)
[Text](#)[Image](#)[Numeric](#)[Audio](#)
- 2) _____ is used to connect to communication network in order to disseminate information. (1)
[CPU](#)[ALU](#)[UNI](#)[U](#)[None of these](#)
- 3) Microphone is an example of _____ device. (1)
[Output](#)[Storage](#)[Input](#)[Processing](#)
- 4) Which of the following is not the characteristic of the Archival Store? (1)
[Non Volatile](#)[Long Life](#)[Portable](#)[None of these](#)
- 5) A parallelogram represents _____ in flow chart. (1)
[Input](#)[Output](#)[Process](#)[\(1\) and \(2\) both](#)
- 6) _____ storage cell is used for laser disks or compact disks. (1)
[Flipflop](#)[Capacitor](#)[Magnetic](#)[None of these](#)
- 7) _____ is used as cache memory in computer system. (1)
[SRAM](#)[DRAM](#)[Both \(1\) and \(2\)](#)[None of these](#)
- 8) The area required to fabricate capacitor storage cell is _____. (1)
[0.02 cm²](#)[0.02 mm²](#)[0.02 nm²](#)[0.02 μm²](#)
- 9) EDI stands for _____. (1)
[Electronic Digital Information](#) [Electronic Data Information](#) [Electric Data Interchange](#) [Electronic Data Interchange](#)
- 10) Communication between Bluetooth devices happens over short-range, ad hoc networks known as _____. (1)
[Adnet](#)[Bnet](#)[Piconet](#)[Manet](#)
- 11) 2's complement of 1011 is _____. (1)
[0100011001011010](#)
- 12) Hexadecimal value of decimal number 679 is _____. (1)

[2B72C721072A7](#)

- 13) Binary division of 11000110 / 110 is _____. (1)
[100001100000110001100000](#)
- 14) _____ is the 9's complement of 3456. (1)
[654465456543None of these](#)
- 15) _____ gate is an electronic circuit that gives a high output (1) if one or more of its inputs are high. (1)
[NANDNOROR\(1\) and \(3\) Both](#)
- 16) Which kind of circuits do not have any memory? (1)
[SequentialCombinational\(1\) and \(2\) bothNone of these](#)
- 17) The value of x.x boolean algebra is _____. (1)
[10xNone of these](#)
- 18) _____ is a set of exactly 2^m adjacent cells containing 1s. (1)
[Product TermKarnaugh MapSub CubeNone of these](#)
- 19) _____ receives one input and distributes it over several outputs. (1)
[AdderMultiplexerDemultiplexerFlipflop](#)
- 20) The flipflop has _____ output signals. (1)
[OneTwoThreeNone of these](#)

SECTION-II

Answer all the questions.

Make suitable assumptions when necessary.

Simple Calculator is permitted.

- 2) Explain computer as Data Processing Model. (5)
Q.2 A)
- [OR] List out of characteristics of storage cell. Explain flipflop storage cell. (5)
Q.2 B)
- Q.2 C) Define Information Technology. Give difference between Data and Information. (3)
- [OR] What is the purpose of RAM? Differentiate SRAM and DRAM. (3)
Q.2 D)
- Q.2 E) Explain usage and applications of GPS. (4)
- [OR] Explain any four emerging trends of Information Technology. (4)
Q.2 F)
- 3) List and explain types of ROM. (5)
Q.3 A)
- Q.3 B) List types of Disk Interface Controller. Explain the usage of disk interface controller. (5)
- [OR] Explain polycarbonate storage cell. (5)
Q.3 C)

Q.3 D) Define following terms : (3)

(i) Interrupt (ii) Data Security (iii) Flow chart

[OR] List any three Business to Customer(B2C) E-commerce transactions. (3)

Q.3 E)

SECTION-III

Answer all the questions.

Make suitable assumptions when necessary.

Simple Calculator is permitted.

4) Perform binary addition of following : (4)

Q.4 A) **(1)** 11001110011 + 11001100111 **(2)** 1111111000011 + 110011100

[OR] Explain NAND and Exclusive OR gates. (4)

Q.4 B)

Q.4 C) Derive the equation from the function $F(A,B,C) = \sum(m_0, m_2, m_4, m_6, m_7)$. (4)

[OR] Perform the following binary subtraction using 2's complement : (4)

Q.4 D) **(a)** 11001100 - 11100000 **(b)** 11001111 - 10110001

Q.4 E) Write De Morgan's rules for boolean algebra and prove them with truth table. (4)

[OR] Convert following decimal numbers to binary : (4)

Q.4 F) (1) 567 (2) 894

5) Explain Half and Full Adder combinational circuits with block diagram and truth table. (6)

Q.5 A)

[OR] Explain Multiplexer and Demultiplexer with proper example. (6)

Q.5 B)

Q.5 C) Explain characteristics of combinational circuit. (3)

[OR] What do you mean by Karnaugh map? Draw the Karnaugh Map for any four variables equation. (3)

Q.5 D)

Q.5 E) Do as directed : (4)

(1) Prove the boolean algebra expression $(x+y)(x+z) = x + yz$ with boolean algebra rules.

(2) Perform Binary Multiplication : 11001 * 101

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